Dv Guidance

Component Design Document

1 Description

Delta-V guidance. Produces the inertially fixed attitude reference for a delta-V burn. The burn frame has its third axis along the commanded delta-V and its second axis as close as possible to a preferred direction, both sent by the state manager when a burn is commanded, so the reference attitude is a constant between burn commands and the reference rate and acceleration are zero. A rotation about the burn axis, when wanted, is applied by a downstream reference rotation taking this reference as its base. Wraps the DvGuidanceAlgorithm C++ algorithm via its C shim.

2 Requirements

No requirements have been specified for this component.

3 Design

3.1 At a Glance

Below is a list of useful parameters and statistics that give a quick look into the makeup of the component.

- **Execution** - *passive*
- **Number of Connectors** - 2
- **Number of Invokee Connectors** - 2
- **Number of Invoker Connectors** - *None*
- **Number of Generic Connectors** - *None*
- **Number of Generic Types** - *None*
- **Number of Unconstrained Arrayed Connectors** - *None*
- **Number of Commands** - *None*
- **Number of Parameters** - *None*
- **Number of Events** - *None*
- **Number of Faults** - *None*
- **Number of Data Products** - *None*
- **Number of Data Dependencies** - *None*
- **Number of Packets** - *None*

3.2 Diagram

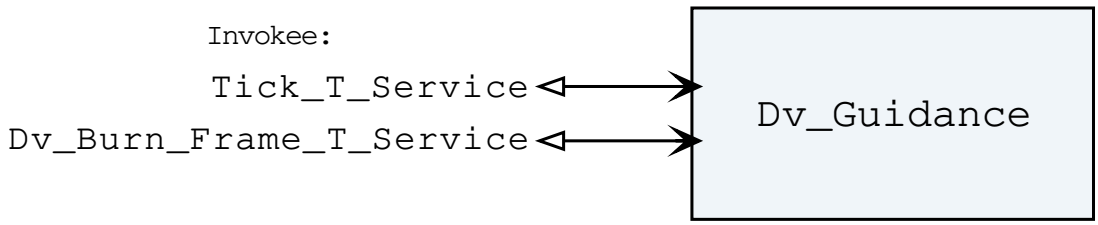


Figure 1: Dv Guidance component diagram.

3.3 Connectors

Below are tables listing the component's connectors.

3.3.1 Invokee Connectors

The following is a list of the component's *invokee* connectors:

Table 1: Dv Guidance Invokee Connectors

Name	Kind	Type	Return_Type	Count
Tick_T_Service	service	Tick.T	Att_Ref.T	1
Dv_Burn_Frame_T_Service	service	Dv_Burn_Frame.T	Packed_Command_Execution_Status.T	1

Connector Descriptions:

- **Tick_T_Service** - Run the algorithm up to the current time and return the attitude reference it produces.
- **Dv_Burn_Frame_T_Service** - Set the burn frame from the commanded delta-V and the preferred direction. The algorithm validates them and returns Success when the frame is applied, or Failure when it rejects them, in which case the previous frame is kept, so the sender can reject its command.

3.3.2 Invoker Connectors

None

3.4 Interrupts

This component contains no interrupts.

3.5 Initialization

Below are details on how the component should be initialized in an assembly.

3.5.1 Component Instantiation

This component contains no instantiation parameters in its discriminant.

3.5.2 Component Base Initialization

This component contains no base class initialization, meaning there is no `init_Base` subprogram for this component.

3.5.3 Component Set ID Bases

This component contains no commands, events, packets, faults or data products that need base identifiers.

3.5.4 Component Map Data Dependencies

This component contains no data dependencies.

3.5.5 Component Implementation Initialization

The calling of this implementation class initialization procedure is mandatory. Initializes the delta-V guidance algorithm with a stand-in burn frame, which the first burn command replaces. The `init` subprogram takes no parameters.

3.6 Commands

The Dv Guidance component has no commands.

3.7 Parameters

The Dv Guidance component has no parameters.

3.8 Events

The Dv Guidance component has no events.

3.9 Data Products

The Dv Guidance component has no data products.

3.10 Data Dependencies

The Dv Guidance component has no data dependencies.

3.11 Packets

The Dv Guidance component has no packets.

3.12 Faults

The Dv Guidance component has no faults.

4 Unit Tests

The following section describes the unit test suites written to test the component.

4.1 *Dv_Guidance_Tests* Test Suite

This is a unit test suite for the Dv Guidance component

Test Descriptions:

- **`Test`** - Check the reference returned for the burn frame of the algorithm's Python reference test, with zero reference rates, to ensure the Ada to C to C++ integration is sound, and that the stand-in frame serves before a burn is commanded.
- **`Test_Burn_Frame_Update`** - Ensure a newly commanded delta-V and preferred axis reach the reference on the next tick.
- **`Test_Invalid_Burn_Frame`** - Ensure a burn frame the algorithm rejects is refused with Failure and leaves the reference unchanged.

5 Appendix

5.1 Preamble

This component contains no preamble code.

5.2 Packed Types

The following section outlines any complex data types used in the component in alphabetical order. This includes packed records and packed arrays that might be used as connector types, command arguments, event parameters, etc..

Att_Ref.T:

Table 2: Att_Ref Packed Record : 288 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Sigma_Rn	Packed_F32x3.T	-	96	0	95
Omega_Rn_N	Packed_F32x3.T	-	96	96	191
Domega_Rn_N	Packed_F32x3.T	-	96	192	287

Field Descriptions:
- **`Sigma_Rn`** - *No description provided.*
- **`Omega_Rn_N`** - *No description provided.*
- **`Domega_Rn_N`** - *No description provided.*

Dv_Burn_Frame.T:

The burn frame of a delta-V burn, as the delta-V guidance takes it when a burn is commanded.

Table 3: Dv_Burn_Frame Packed Record : 192 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Cmd_Dv_N	Packed_F32x3.T	-	96	0	95
R_Hat_Pref_N	Packed_F32x3.T	-	96	96	191

Field Descriptions:
- **`Cmd_Dv_N`** - [m/s] Commanded delta-V in inertial frame components. The third axis of the burn frame points along it. Must be finite and at least 1.0E-6 m/s long.
- **`R_Hat_Pref_N`** - [-] Preferred direction of the second axis of the burn frame in inertial frame components. Must be finite, within 1.0E-3 of unit length, and not within about 0.06 degrees

of parallel or antiparallel to the commanded delta-V. Only its component perpendicular to the delta-V shapes the frame.

Packed_Command_Execution_Status.T:

A packed record for holding the command execution status.

Table 4: Packed_Command_Execution_Status Packed Record : 8 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Status	Command_Enums. Command_Execution_ Status.E	0 => Success 1 => Failure	8	0	7

Field Descriptions:

- **Status** - The command execution status.

Packed_F32x3.T:

Packed 3 element vector of 32-bit floats.

Table 5: Packed_F32x3 Packed Array : 96 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Short_Float	-3.40282e+38 to 3.40282e+38	32	3	96

Sys_Time.T:

A record which holds a time stamp using EMA VTC format including seconds and subseconds since epoch (1-5-1980 to 1-6-1980 midnight).

Table 6: Sys_Time Packed Record : 48 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Seconds	Interfaces. Unsigned_32	0 to 4294967295	32	0	31
Subseconds	Interfaces. Unsigned_16	0 to 65535	16	32	47

Field Descriptions:

- **Seconds** - The number of seconds elapsed since epoch.
- **Subseconds** - The number of 1/(2^16) sub-seconds.

Tick.T:

The tick datatype used for periodic scheduling. Included in this type is the Time associated with a tick and a count.

Table 7: Tick Packed Record : 80 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	48	0	47
Count	Interfaces. Unsigned_32	0 to 4294967295	32	48	79

Field Descriptions:

- **Time** - The timestamp associated with the tick.
- **Count** - The cycle number of the tick.

5.3 Enumerations

The following section outlines any enumerations used in the component.

Command_Enums.Command_Execution_Status.E:

This status enumeration provides information on the success/failure of command execution, as returned by the command handler.

Table 8: Command_Execution_Status Literals:

Name	Value	Description
Success	0	Command was successfully executed.
Failure	1	Command was not successfully executed.